

Making Emergency Care Interpretable: Journey-Level Information Breakdowns from Patient and Caregiver Perspectives

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Abstract

Emergency departments (EDs) are highly dynamic and uncertain care environments in which patients struggle to understand their current status, what may happen next, and the reasons behind waiting and care progression. This work presents an exploratory qualitative study examining how information-related challenges unfold across patients' ED journeys. Through semi-structured interviews with patients ($n = 5$) and caregivers ($n = 7$) and inductive thematic analysis, we identify three recurring information breakdowns across ED stages: limited process legibility and guidance, uninterpretable waiting, and a lack of understandable intermediate explanations. We introduce a patient-facing conceptual prototype that illustrates how existing operational states could be translated into patient-comprehensible updates. Designed as a low-burden, illustrative concept, the prototype aims to support patients' and caregivers' sensemaking of ED care progression without introducing additional clinical workload. Together, these breakdowns informed corresponding design modules in our prototype, clarifying how journey-level information support can be systematically translated into patient-facing design.

CCS Concepts

• Health informatics; • User studies; • Empirical studies in HCI;

Keywords

Journey-level information breakdowns, Patient-Facing Design, Information Transparency, Waiting and Uncertainty, Healthcare Sense-making, Interpretable operational states, Emergency Care

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1 Introduction

For many patients and caregivers, an emergency department (ED) visit begins abruptly, often without preparation, and quickly becomes an unpleasant experience of waiting, fragmented updates, and uncertainty about their current situation and upcoming procedures. From the hospital's perspective, however, EDs operate under continual reprioritization, unexpected arrivals, and resource reallocation, making care sequences and timing uncertain [12]. This mismatch between patients' need for clarity and hospitals' need for flexibility makes communicating and managing uncertainty central to patient experience and safety, particularly when diagnoses remain provisional or when observation and iterative assessment are required [16].

These conditions foreground challenges of *information transparency*: whether patients and accompanying caregivers can access, interpret, and act upon information about their immediate status and the anticipated care trajectory. Prior work has examined ED touchpoints but often provides a fragmented view of how information is encountered or disrupted across a visit [3]. Yet patient-centered accounts that characterize information breakdowns across ED stages remain limited.

ED care also involves in-progress information and provisional plans; for patients and caregivers, limited update and opaque processes can impede sensemaking. To address this gap, we conducted semi-structured interviews with patients ($n = 5$) and caregivers ($n = 7$) and performed thematic analysis. Our findings identify recurring journey-level information breakdowns and demonstrating how an information transparency lens can inform patient-facing designs. This work specifically focuses on articulating journey-level information breakdowns and their corresponding design implications for emergency care.

1.1 Patients'/Caregivers' Uncertainty and Sensemaking during ED Journey

Prior ED research has extensively examined system efficiency and performance metrics, yet fewer patient-centered accounts examine how people interpret information across the ED journey and how confusion and uncertainty persist across stages. [4, 7]. When diagnostic or treatment uncertainty is poorly communicated, patients' understanding, trust, and agency can be undermined [6]. In EDs, uncertainty stems not only from indeterminate outcomes but also from gaps in shared understanding during discussions of tests,

treatments, and next steps [14]. With limited or fragmented informational cues, patients and caregivers often engage in sensemaking to interpret unfolding events and prepare for future outcomes. [6, 11]. Taken together, these conditions highlight how ED experiences are fragmented across interactions and stages, and suggest the need for journey-oriented information support that helps patients and caregivers navigate care progression over time rather than relying on isolated updates.

1.2 Patient-Facing Information Support

As patients and caregivers navigate uncertainty in ED care, information and communication become key sources of guidance. However, prior works suggest that transparency alone does not guarantee comprehension [11, 14]. In ED settings, perceived communication quality is closely tied to trust and care evaluations [5]. Studies of patient portals and access to test results further show that patients may struggle to interpret information due to its complexity and limited context. This difficulty can, in turn, increase anxiety [18, 19, 22, 23].

Human-centered design highlights opportunities for patient-facing tools that translate complex clinical experiences into actionable informational support grounded in patients' lived experiences [10]. In CHI, researchers have examined emergency patients' adaptation to information challenges and the design of patient-friendly medical explanations [15, 20], yet this work primarily focuses on discrete interactions rather than journey-level sensemaking across stages. Previous studies related to the ED and the digital pathway demonstrate the benefits of informational interventions during waiting periods and care transitions, but these efforts often focus on isolated tools or moments [2, 9, 17]. What remains underexplored is how information breakdowns unfold across the *entire* ED journey and how they can be translated into integrated, cross-stage design opportunities from the patient perspective.

2 Method

2.1 Study Design

We conducted an exploratory qualitative study using semi-structured interviews to examine how patients and accompanying caregivers experienced ED journeys and made sense of events across care stages. This format allowed participants to recount their experiences in their own words while enabling the interviewer to probe key moments, transitions, and interactional challenges as they emerged. Interviews focused on participants' perceptions, interpretations, and emotional responses throughout the ED process.

2.2 Participants and Recruitment

We interviewed $N = 12$ adults in Taiwan with previous experience in emergency department, including 5 patients and 7 accompanying caregivers (Table 1). Interviews were conducted outside clinical settings. Participants were recruited via a public Taiwanese research-participation Facebook group using convenience sampling and were eligible if they were ≥ 18 years old, able to provide informed consent and sufficiently proficient in Mandarin. We did not restrict participation by medical condition or hospital type. Time since the ED visit ranged from 3 days to 3 years, capturing both recent and retrospective accounts.

2.3 Study Procedure

Interviews were conducted remotely via Google Meet and lasted approximately 60 minutes each. Participants first completed a brief recruitment survey to assess eligibility and availability, after which interviews were scheduled. Prior to the interview, the researcher explained the study and obtained informed consent.

The interview guide followed a journey-oriented structure, prompting participants to reflect on their experiences across key ED stages, spanning *Before Arrival*, *Triage*, *Waiting*, *Medical Consultation*, *Examination and Treatment*, *Observation & Assessment*, *Medical Decision*, and *Leaving the ED*. For each stage, participants were asked to describe what occurred, what information and communication they received, moments of confusion or difficulty, and how they interpreted what was happening and what might happen next.

With permission, interviews were audio-recorded and transcribed verbatim. The study was Institutional Review Board approved (November 2025).

2.4 Data Analysis

We used inductive thematic analysis. The two authors iteratively read transcripts, generated initial codes, and discussed discrepancies to refine the coding scheme before clustering codes into higher-level themes. ATLAS.ti was used to support coding, theme development, and retrieval of illustrative excerpts. This analytic process focused on identifying recurring experiential breakdowns and interactional tensions across the dataset. Themes were considered robust when they recurred across multiple interviews, including both patients and caregivers, and no substantially new breakdown types emerged in later interviews.

3 Results

Through inductive analysis, we identified recurring journey-level information breakdowns, which we analyze below through two conditions: *information transparency*, referring to whether information is visible, timely, and interpretable; and *information asymmetry*, referring to unequal access to or interpretation of information between clinicians and patients. Figure 1 summarizes these breakdowns along the ED journey. Collectively, they shape patients' ongoing sensemaking across the ED journey.

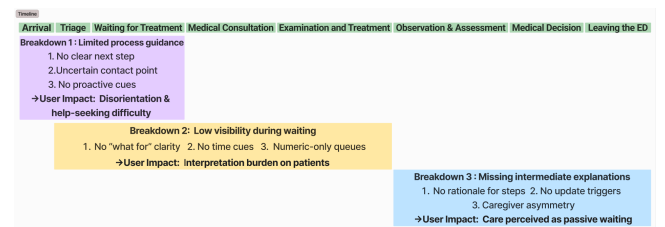


Figure 1: Journey-Level information breakdowns: (1) Limited process guidance at arrival; (2) Low visibility into waiting status; and (3) Insufficient intermediate explanations for in-progress care.

ID	Role	Demographics	Chief Complaint (Medical Context)
P1	Patient	Male, 18-30	Finger injury (Fracture & Laceration)
P2	Caregiver	Female, 35-40	Severe headache (Intracerebral Hemorrhage)
P3	Caregiver	Male, 30-35	Fever & Dyspnea (Pediatric Infection)
P4	Patient	Female, 18-30	Worsening wound pain (Cellulitis)
P5	Caregiver	Female, 18-30	Severe abdominal pain (Gallstones)
P6	Patient	Male, 31-45	Dyspnea (Pleural Effusion)
P7	Caregiver	Female, 18-30	Sudden fainting (Syncope)
P8	Caregiver	Female, 18-30	Blood in urine (Hematuria)
P9	Caregiver	Female, 31-45	Infectious symptoms (COVID-19)
P10	Caregiver	Female, 18-30	Abdominal pain (Chronic Complications)
P11	Patient	Female, 18-30	Acute abdominal pain (Duodenal Perforation)
P12	Patient	Female, 18-30	Facial deformity (Nasal Fracture)

Table 1: Participant Demographics.

3.1 Breakdown 1: Limited process guidance undermines help-seeking and next-step clarity

In *Before Arrival*, *Triage*, and early *Waiting*, participants sought actionable cues for process guidance, such as who to approach, where to go, and what happens next. When such guidance was not visible or actionable, participants described difficulty initiating help-seeking and uncertainty about appropriate points of contact. For example, P10 said “I didn’t know who is in charge of my mother’s care... so I didn’t know who to call for help.”

This breakdown reflects low information transparency around roles and points of contact, which in turn produced information asymmetry by requiring patients and caregivers to infer how the ED system operated. In contrast, proactive entry guidance and lightweight wayfinding cues (e.g., floor lines) made the process feel legible and reassuring. As P8 described “There was someone at the entrance who guided us... the whole process was very clear, so I felt reassured.”

3.2 Breakdown 2: Low visibility into waiting produces “waiting without knowing what for”

Breakdown 2 was most salient during *Waiting* and *Observation & Assessment*. Participants emphasized that distress stemmed less from prolonged waits than from limited visibility into what they were waiting for or when progress might occur. While queue displays conveyed ordering, they offered little interpretive support. As P3 noted “There was an electronic queue board ... but no time information. We could only guess based on the numbers.”

Here, information transparency was reduced to numerical indicators, placing the burden of interpretation on patients and caregivers. Participants suggested that even rough estimates or partial updates (e.g., what is pending and when updates may occur) would help align expectations and reduce anxiety under uncertainty. As P12 proposed: “Maybe instead of just saying ‘wait a bit,’ they could say roughly when the results might come out.”

3.3 Breakdown 3: Missing intermediate explanations obscure the rationale for care decisions

Breakdown 3 recurred from *Consultation* through *Medical Decision*. When tests, treatments, or observation were not promptly explained, participants interpreted care as passive waiting and questioned its necessity. As P5 illustrated “I was just lying there ... for four hours without any treatment ... So why was I kept in that room?”

Participants did not seek definitive diagnoses; instead, they sought *intermediate understanding*: high-level explanations of what clinicians were considering, what steps were planned, and what would trigger updates. Caregivers, particularly when spatially separated from patients, described heightened information asymmetry that further limited their ability to make sense of ongoing care. As P6 recounted: “My brother sat outside for six hours, not knowing what was happening at all.”

4 Discussion

4.1 Theoretical Implications

This study contributes to HCI understandings of information transparency and information support by showing that information-related difficulties in ED care are not isolated breakdowns, but recur across stages and shape ongoing sensemaking throughout the care journey. These conditions hindered the ability of the participants to construct a coherent narrative of what was happening and why, highlighting the temporal and narrative dimensions of information transparency.

Extending prior HCI and healthcare research that often focuses on discrete touchpoints [1, 8, 13, 21], our findings foreground information transparency as a *journey-level and temporal property* of emergency care. We shift attention toward *interpretable states* and *explicit update triggers*. This reframing advances a design stance for HCI that treats information transparency not as eliminating uncertainty, but as making uncertainty interpretable and navigable: an insight that extends beyond emergency care to other dynamic, time-pressured human–system interactions.

4.2 Design Implications

We translate the identified breakdowns into a patient-facing prototype concept, designed as a QR-code-based ED journey status page accessible on patients' own devices (Figure 2). This interface integrates three complementary design modules—journey legibility, interpretable waiting, and intermediate explanations—by rendering existing operational states into patient-comprehensible updates. To ensure feasibility and minimize clinical workload, the prototype reuses existing operational data to populate standardized templates, maintaining low-burden implementation for clinical staff. Users authenticate using patient-controlled information (e.g., date of birth) and may optionally share access with caregivers.

4.2.1 Module 1: Journey legibility and next steps. In early stages of emergency care, patients and caregivers seek immediately actionable cues (e.g., current stage, next steps, and points of contact). When such cues are absent, help-seeking costs rise and uncertainty escalates. To support early-stage orientation and help-seeking, the interface should communicate the current ED stage (e.g., triage completed, awaiting clinician review or test results) alongside likely next steps using non-committal language. By foregrounding actionable cues, such as what is happening now and who may be involved, the design reduces the cost of initiating help-seeking without overpromising timelines.

4.2.2 Module 2: Interpretable waiting states. According to our results, distress during waiting stemmed less from duration than from uncertainty about what one is waiting for. Instead of promising precise time estimates, designs should communicate interpretable waiting states (e.g., sample received, in progress, pending tests or reviews), simplified status labels, and explicit update triggers to help manage expectations under uncertainty, aligning with the dynamic and reprioritized nature of ED workflows.

4.2.3 Intermediate explanations for in-progress care. Our results suggest that participants did not expect immediate diagnoses, but wanted to understand why actions were taken, what clinicians were considering, and when updates might occur. Brief, patient-comprehensible intermediate explanations provide high-level rationales for tests, treatments, or observation, clarify what clinicians may be considering, and indicate when updates are likely to occur. These intermediate explanations can reduce anxiety and mitigate information gaps, particularly when patients and caregivers are spatially separated.

4.3 Limitations and Future Directions

This study has several limitations. First, participants were recruited online and reflected retrospectively on prior ED visits; accounts may be subject to recall bias and may not represent all patient populations or institutional workflows. Nevertheless, our analysis focused on recurring patterns of information experience rather than precise clinical sequences; therefore, retrospective accounts were sufficient to identify journey-level information breakdowns. Second, the study is situated in Taiwanese emergency care, and transferability to other healthcare systems and cultural contexts warrants further examination. Third, the proposed prototype is conceptual and low-fidelity; as a result, it does not provide evidence of usability, effectiveness, or feasibility in clinical practice.

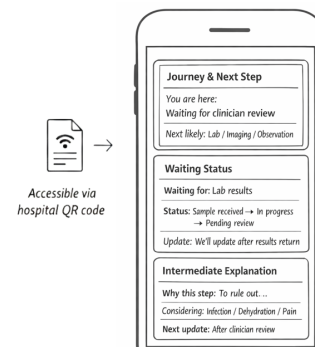


Figure 2: Conceptual low-fidelity prototype illustrating how existing operational states could be translated into patient-comprehensible updates across the ED journey. The status page presents three modules: journey stage and next steps, interpretable waiting states, and intermediate explanations with update triggers.

Future work may examine journey-level information transparency in situ, focusing on how patients, caregivers, and clinicians perceive and make sense of interpretable states and update triggers in emergency care. Such work could also explore how journey-level information representations align with clinicians' existing information practices, without introducing additional documentation or coordination burdens. Future studies may further investigate situations in which such cues fail to support sensemaking, such as during rapid reprioritization or unexpected clinical escalation. Longer-term and cross-context studies could additionally examine how journey-level informational cues support sensemaking under uncertainty across different healthcare systems and other high-stakes service environments.

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